Edge rate limits & observability (Agent 0)

YaPcore ships **connection flood protection** on YaP Link, a **lag-machine governor** plugin, and **Prometheus** `/metrics` scrape endpoints — without forking Folia.

Operator playbook (public edge): EDGE_HARDEN.md

YaP Link rate limits (default ON)

Keys in `link.properties` / `link.toml` (`examples/yap-link/`, `link-data/`):

Key	Default	Meaning
`connect-rate-limit-enabled`	`true`	Per-IP TCP accept throttle
`connect-rate-per-ip`	`20`	Max accepts / window
`connect-rate-window-ms`	`10000`	Window length
`handshake-rate-limit-enabled`	`true`	Handshake packet flood drop
`handshake-rate-per-ip`	`40`	
`handshake-rate-window-ms`	`10000`	
`login-rate-limit-enabled`	`true`	Login-intent flood drop
`login-rate-per-ip`	`10`	
`login-rate-window-ms`	`10000`	
`rate-limit-exempt-loopback`	`true`	Skip limits for `127.0.0.1` / `::1`
`max-concurrent-per-ip-enabled`	`true`	Cap simultaneous TCP sessions / IP
`max-concurrent-per-ip`	`8`	`0` disables the cap

Loopback is exempt by default so smoke scripts and local ops are not tripped. Soak with `rate-limit-exempt-loopback=false` (see script below).

Public vs LAN

Profile	connect/ip	concurrent/ip	Notes
Public	`20` / 10s	`8`	Defaults
Under attack	`8` / 10s	`4`	Fail-closed — EDGE_HARDEN.md
LAN / NAT	`100` / 10s	`32`	Many clients one IP

Drops increment: `connect.throttled`, `handshake.dropped`, `login.dropped`, `connect.concurrent_dropped`.

Scrape (clear names):

```
yap_link_connect_throttled_total
yap_link_handshake_dropped_total
yap_link_login_dropped_total
yap_link_connect_concurrent_dropped_total
yap_link_throttle_*_drops   # gauge mirrors for dashboards
```

Prove:

```
```

YaPLagGuard

See LAGGUARD.md. Jar via `gradle installProductDefaults` / `assembleRelease`.

Survival-oriented defaults: entities/chunk `72`, primed TNT `8`, hopper `48`/20t, redstone `96`/20t.

```
```

Prometheus scrape

Chassis

`GET http://127.0.0.1:8080/metrics` — **firewall**; no auth.

Exports: players, ticks, heap, ThreadMetrics, `yapcore_lagguard_*`.

YaP Link

```
metrics-http-enabled=true
metrics-http-bind=127.0.0.1
metrics-http-port=9091
```

`GET http://127.0.0.1:9091/metrics` — throttle counters above.

Example Prometheus scrape config

```
scrape_configs:
  - job_name: yapcore
    static_configs:
      - targets: ["127.0.0.1:8080"]
    metrics_path: /metrics
  - job_name: yap-link
    static_configs:
      - targets: ["127.0.0.1:9091"]
    metrics_path: /metrics
```

Dashboard

`GET /api/status` (authed) → `observability` (lagguard + `linkMetricsHint` for throttle metric names).